



US012415893B2

(12) **United States Patent**
Wang et al.

(10) **Patent No.:** **US 12,415,893 B2**

(45) **Date of Patent:** **Sep. 16, 2025**

(54) **ANHYDRIDE COMPOUND, POLYIMIDE, AND THIN FILM**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 636 days.

(21) Appl. No.: **17/888,269**

(22) Filed: **Aug. 15, 2022**

(65) **Prior Publication Data**

US 2023/0101144 A1 Mar. 30, 2023

Related U.S. Application Data

(60) Provisional application No. 63/240,584, filed on Sep. 3, 2021.

(30) **Foreign Application Priority Data**

Dec. 7, 2021 (TW) 110145616

(51) **Int. Cl.**
C08G 73/10 (2006.01)
C07D 307/89 (2006.01)
C08J 5/18 (2006.01)

(52) **U.S. Cl.**
 CPC **C08G 73/1071** (2013.01); **C07D 307/89** (2013.01); **C08G 73/1028** (2013.01); **C08J 5/18** (2013.01); **C08J 2379/08** (2013.01)

(58) **Field of Classification Search**
 None
 See application file for complete search history.

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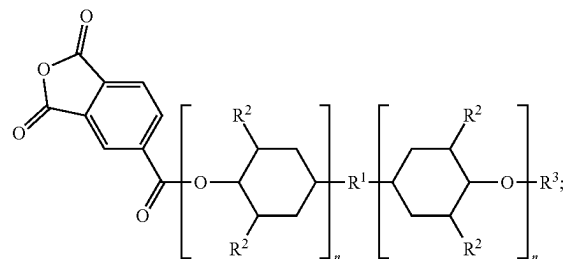
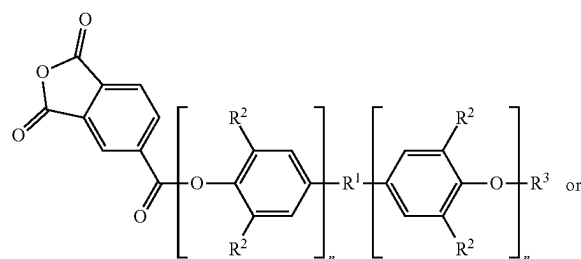
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Primary Examiner — Megan McCulley

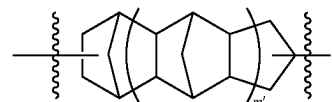
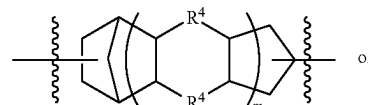
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(57) **ABSTRACT**

An anhydride compound, polyimide, and thin film are provided. The anhydride compound has a chemical structure of

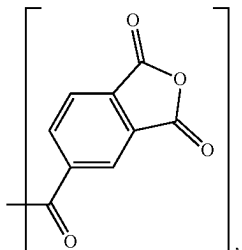


wherein R¹ is



(Continued)

each of R^4 is independently C_{1-6} alkylene group, m is an integer of 0 to 10, and m' is an integer of 1 to 10; n is an integer of 1 to 10, each of R^2 is independently hydrogen, saturated or unsaturated C_{1-6} hydrocarbon group, CF_3 , silanol group, silyl group, or $Al(OH)_3$; and R^3 is



silanol group, silyl group, or $Al(OH)_3$. The anhydride compound can be reacted with a diamine compound to form a polyimide.

9 Claims, No Drawings

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TW Office Action was issued on Jun. 9, 2022 for the corresponding application No. 110145616 in Taiwan.

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1

ANHYDRIDE COMPOUND, POLYIMIDE, AND THIN FILM

CROSS REFERENCE TO RELATED APPLICATIONS

This application claims the benefit of U.S. Provisional Application No. 63/240,584, filed on Sep. 3, 2021, the entirety of which is incorporated by reference herein.

The present application is based on, and claims priority from, Taiwan Application Serial Number 110145616, filed on Dec. 7, 2021, the disclosure of which is hereby incorporated by reference herein in its entirety.

TECHNICAL FIELD

The technical field relates to anhydride compounds.

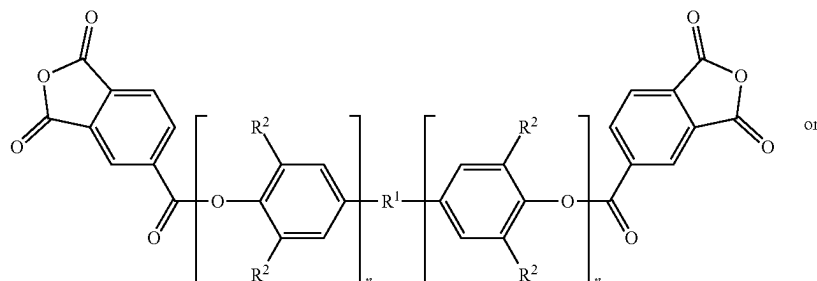
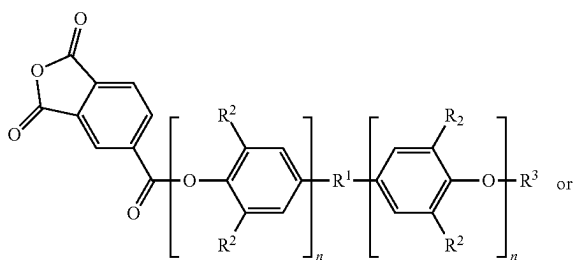
BACKGROUND

Networks are gaining increasing popularity, and consumers are demanding high-quality communication at higher-speeds and lower latency. In response to the 5G era, which promises wider network coverage and faster transmission rates, the use of soft-board materials demands low transmission loss (Dielectric Loss, or Df) and low dielectric coefficients (Dielectric Constant, or Dk) be greatly increased. Using polyimide (PI) as an example, the board material of a conventional polyimide (PI) film cannot satisfy the requirements for high frequencies and high speeds. As such, the modified PI (MPI) for 5G high frequency and high speed applications means that the product is attracting much attention in this industry.

Accordingly, development of a novel synthesis raw material for the board material is called-for to satisfy the above requirements.

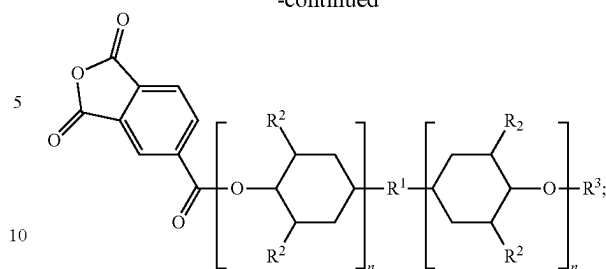
SUMMARY

One embodiment of the disclosure provides an anhydride compound, having a chemical structure of

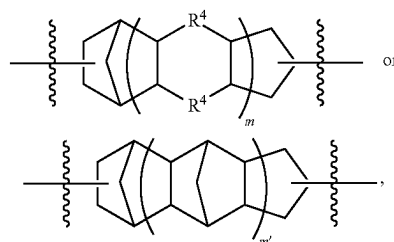


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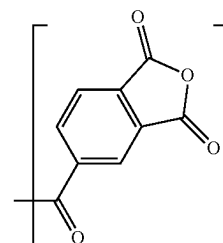
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wherein R¹ is



each of R⁴ is independently C₁₋₆ alkylene group, m is an integer of 0 to 10, and m' is an integer of 1 to 10; n is an integer of 1 to 10; each of R² is independently hydrogen, saturated or unsaturated C₁₋₆ hydrocarbon group, CF₃, silanol group, silyl group, or Al(OH)₃; and R³ is



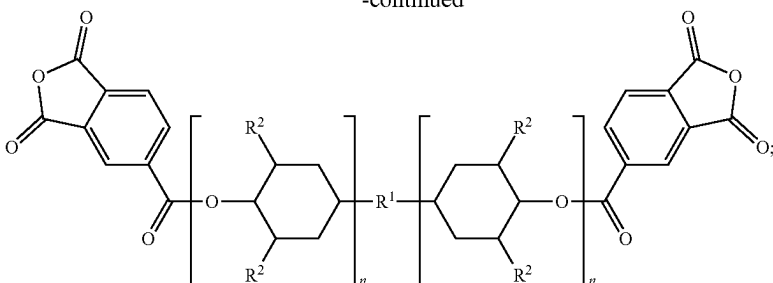
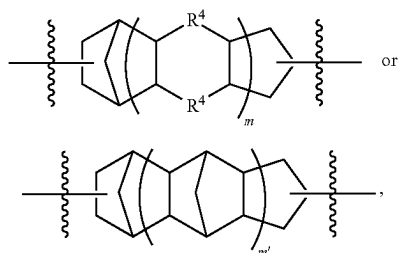
silanol group, silyl group, or Al(OH)₃.

One embodiment of the disclosure provides a polyimide, being formed by reacting an anhydride compound and a diamine compound; wherein the anhydride compound has a chemical structure of

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-continued

wherein R¹ is

each of R⁴ is independently C₁₋₆ alkylene group, m is an integer of 0 to 10, and m' is an integer of 1 to 10; n is an integer of 1 to 10; and each of R² is independently hydrogen, saturated or unsaturated C₁₋₆ hydrocarbon group, CF₃, silanol group, silyl group, or Al(OH)₃.

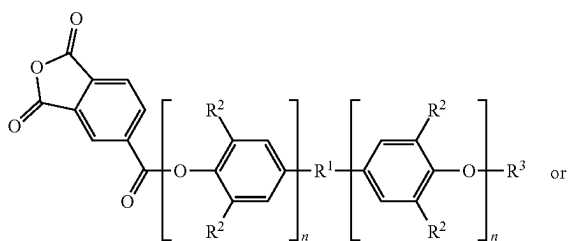
One embodiment of the disclosure provides a thin film, including the described polyimide, wherein the thin film has a thickness of 10 μm to 75 μm, and a dielectric constant of 2.5 to 3 at 10 GHz.

A detailed description is given in the following embodiments.

DETAILED DESCRIPTION

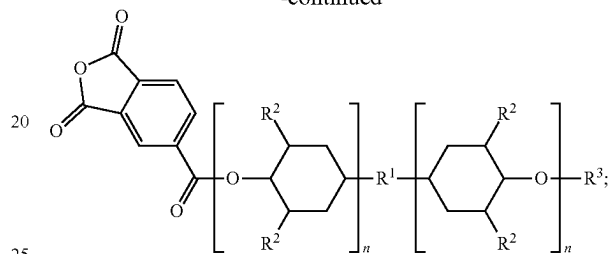
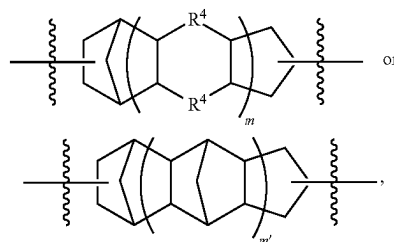
In the following detailed description, for purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of the disclosed embodiments. It will be apparent, however, that one or more embodiments may be practiced without these specific details.

One embodiment of the disclosure provides an anhydride compound, having a chemical structure of



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-continued

wherein R¹ is

each of R⁴ is independently C₁₋₆ alkylene group, m is an integer of 0 to 10, and m' is an integer of 1 to 10; n is an integer of 1 to 10; each of R² is independently hydrogen, saturated or unsaturated C₁₋₆ hydrocarbon group, CF₃, silanol group, silyl group, or Al(OH)₃; and R³ is

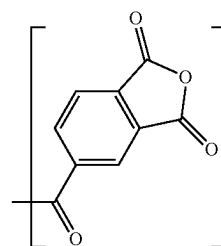
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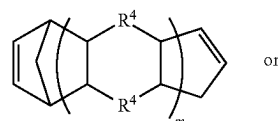
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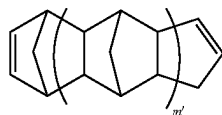


silanol group, silyl group, or Al(OH)₃. The anhydride compound can be synthesized in the following steps:

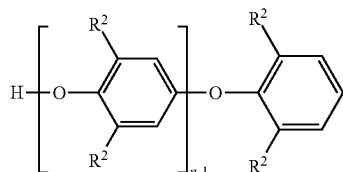


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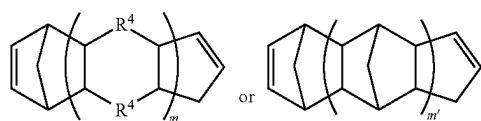
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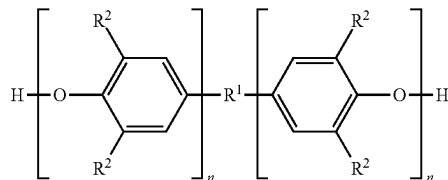
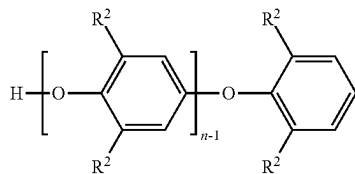
can react with



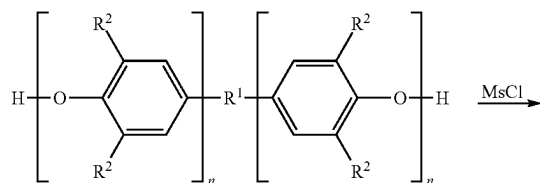
in the presence of acidic catalyst (e.g. AlCl_3 , HF, or another acidic catalyst) to form a diol compound, as shown in below formula.



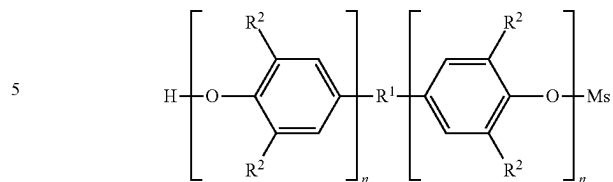
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Subsequently, benzyl bromide (BnBr), chloromethyl methyl ether (MeOCH_2Cl), tert-butyldimethylsilyl ether (TBSCl), methanesulfonyl chloride (MsCl), or dimethyl sulfate is used to protect one alcohol group on one side of the diol compound. Take MsCl as an example, the reaction is shown as below formula.

 $\xrightarrow{\text{MsCl}}$ **6**

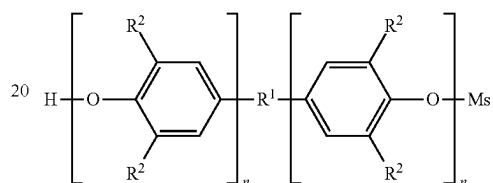
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Subsequently, the compound containing the protection group can react with trimellitic anhydride chloride (TMAC) to form an anhydride compound, as shown in below formula.

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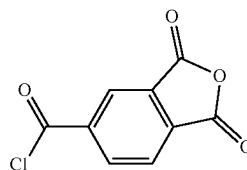


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 $\longrightarrow$

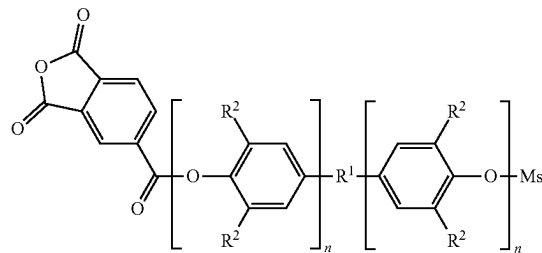
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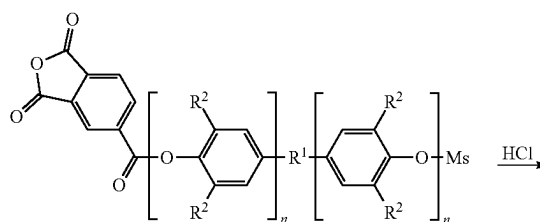
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Subsequently, a de-protection reaction is performed, as shown in below formula. The de-protection reagent can be HCl, I_2 , F^- , LDA (lithium diisopropylamide), BBr_3 , or another suitable reagent.

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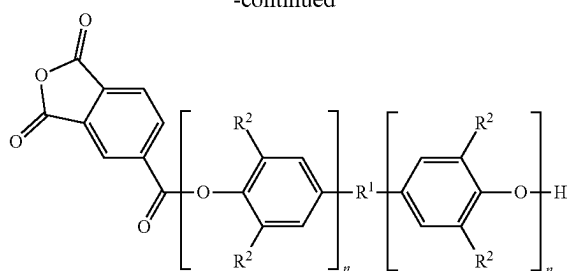
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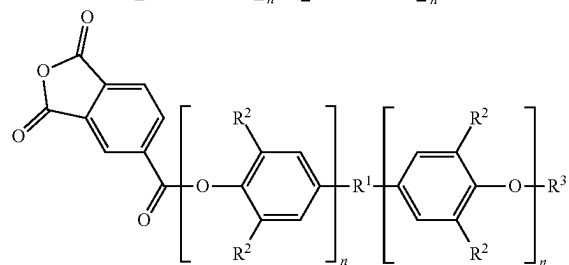
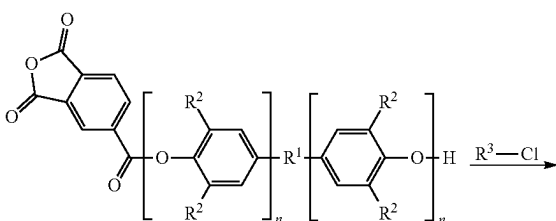
 $\xrightarrow{\text{HCl}}$

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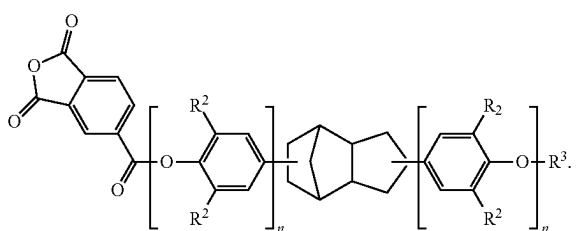
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Subsequently, R^3-Cl can be added to perform a substitution reaction to form an anhydride compound, as shown in below formula.

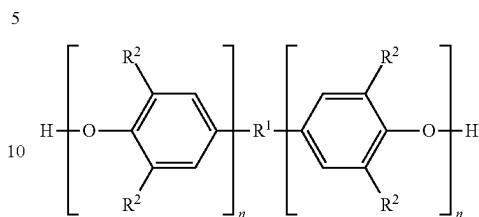


In some embodiments, the anhydride compound formed by the above reactions may have a chemical structure of



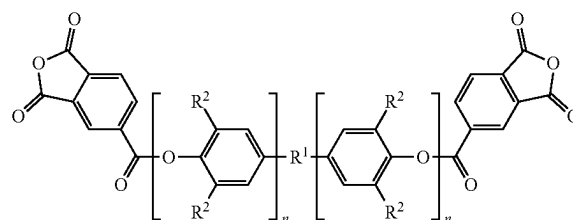
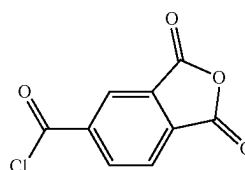
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Alternatively, the diol compound can directly react with TMAC to form an anhydride compound, as shown in below formula.

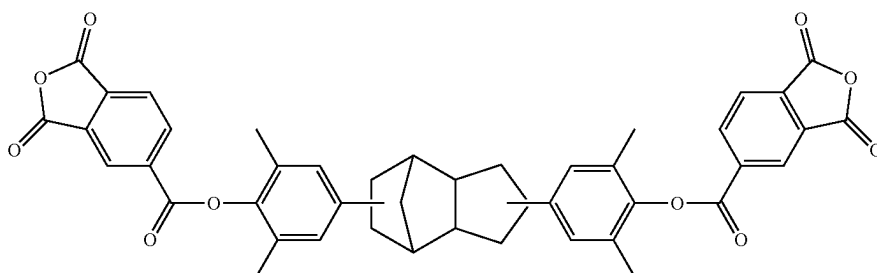


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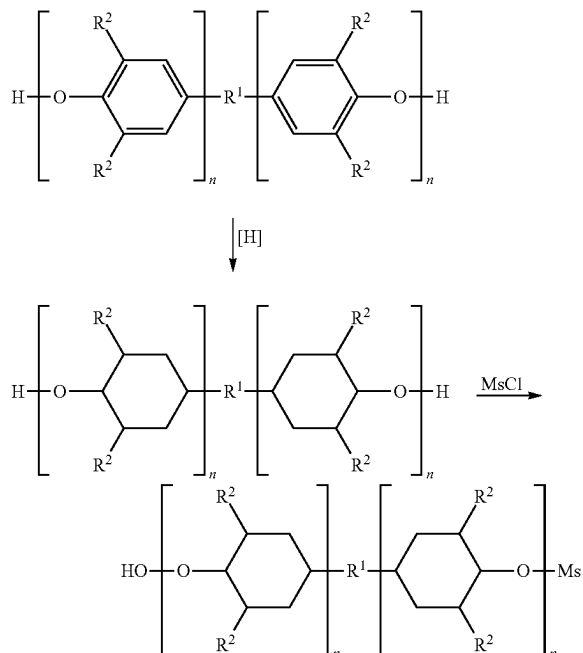


In some embodiments, the anhydride compound formed by the above reactions may have a chemical structure of

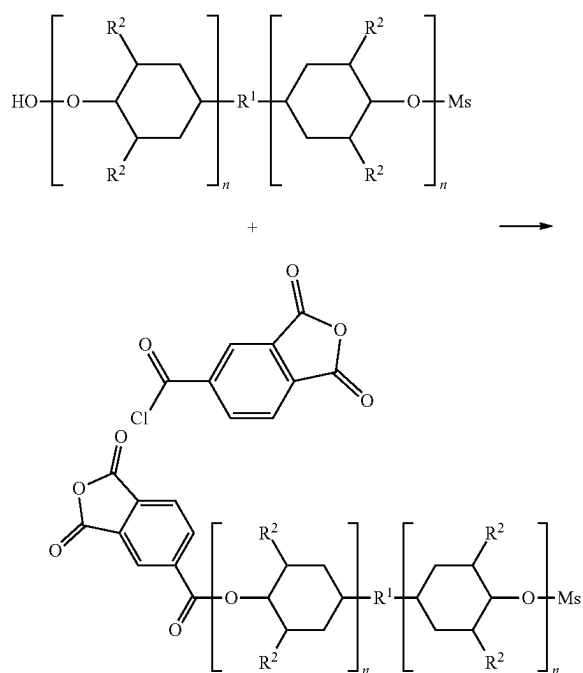


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In some embodiments, the diol compound can be firstly hydrogenated. BnBr, MeOCH₂Cl, TBSCl, MsCl, or dimethyl sulfate is then used to protect one alcohol group on one side of the diol compound. Take MsCl as an example, and the reaction is shown as below formula.

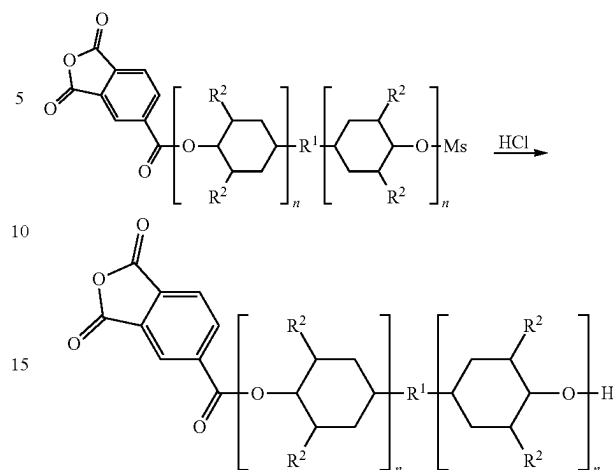


Subsequently, the compound containing the protection group can react with TMAC to form an anhydride compound, as shown in below formula.

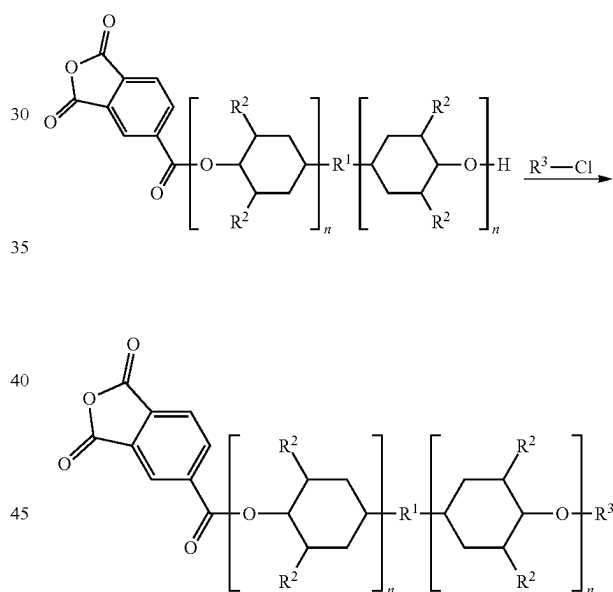


Subsequently, a de-protection reaction is performed, as shown in below formula. The de-protection reagent can be HCl, I₂, F⁻, LDA, BBr₃, or another suitable reagent.

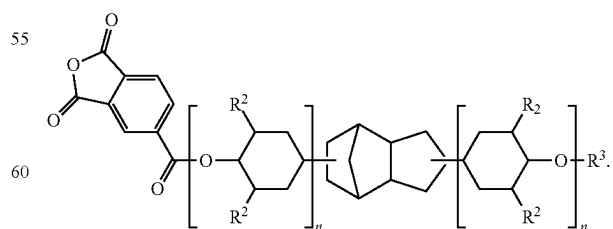
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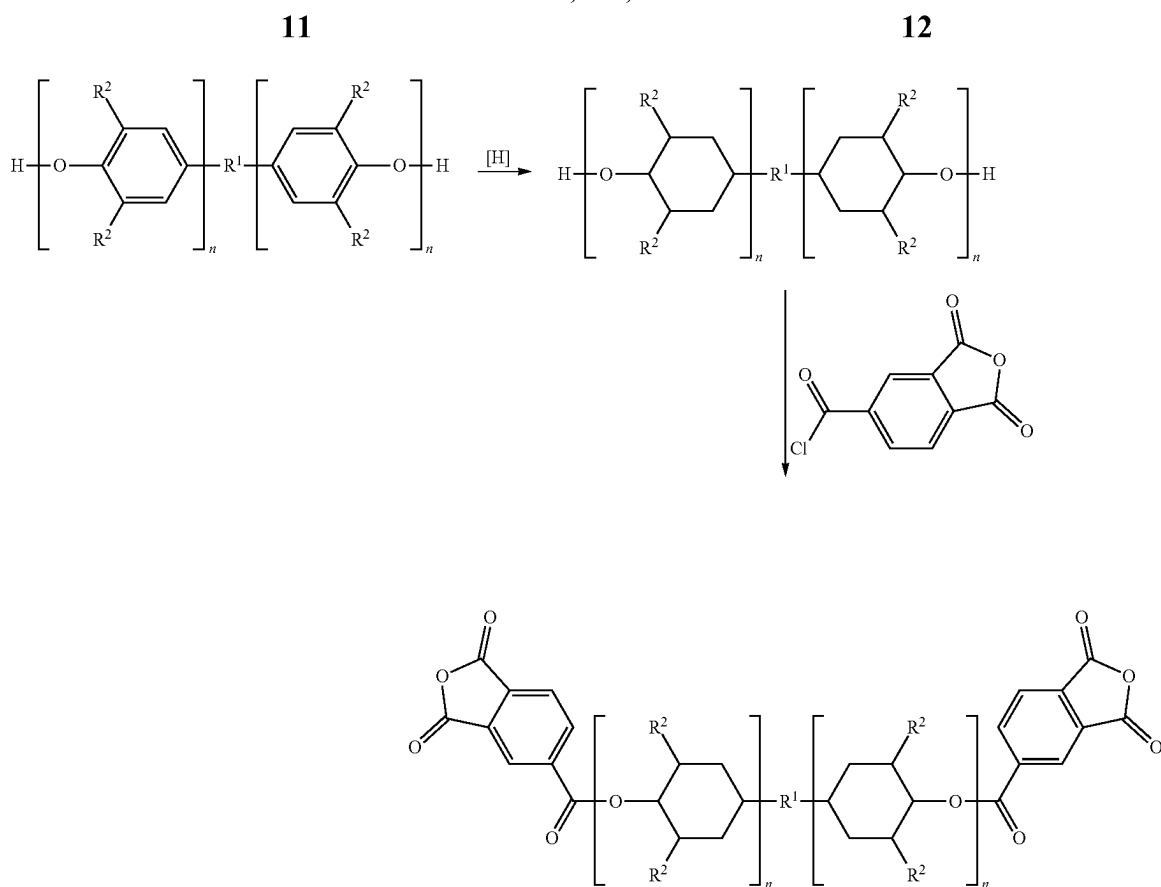
Subsequently, R³-Cl can be added to perform a substitution reaction to form an anhydride compound, as shown in below formula.



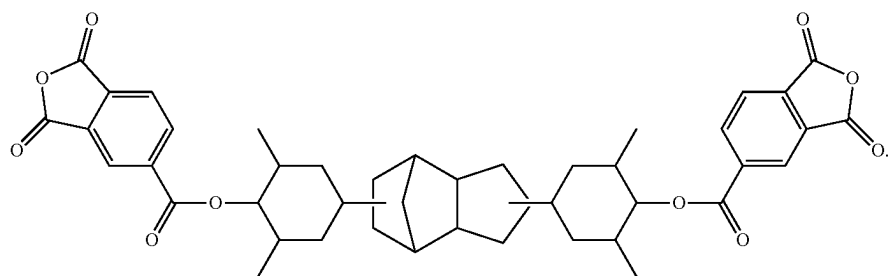
In some embodiments, the anhydride compound formed by the above reactions may have a chemical structure of



On the other hand, the diol compound can firstly undergo hydrogenation, and then react with TMAC to form the anhydride compound, as shown in below formula.



In some embodiments, the anhydride compound formed by the above reactions may have a chemical structure of



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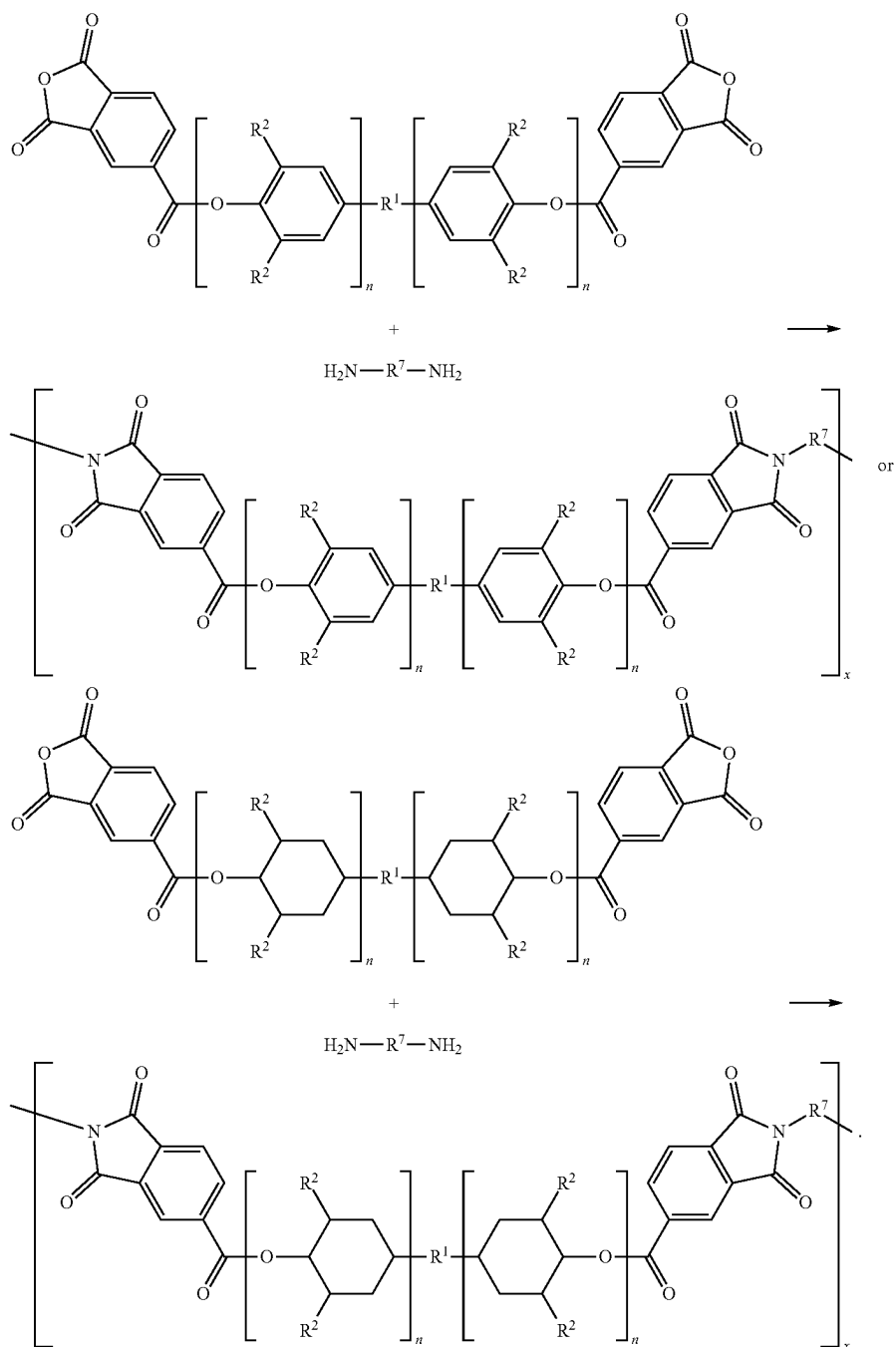
It should be understood that the above reactions are only for illustrating a possible synthesis path of the anhydride compound, which is not the only applicable synthesis path. One skilled in the art may adopt any applicable synthesis path according to his/her requirements for forming the anhydride compound in the disclosure.

One embodiment of the disclosure provides a polyimide formed by reacting the described anhydride compound with a diamine compound. Take dianhydride compound as an example, the reaction is shown below:

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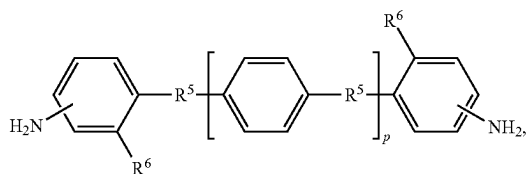
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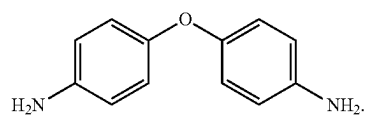


In the above formula, R^7 correspond to the diamine compound type, as described below. x is the repeating number of the repeating unit in the polyimide, which is related to the molecular weight of the polyimide.

In some embodiments, the diamine compound has a chemical structure of

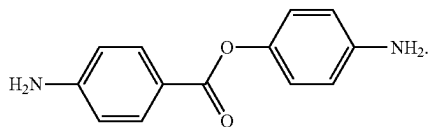


wherein p is an integer of 0 to 5; each of R^5 is independently $-O-$, $-NH-$, C_{1-4} alkylene group, $-SO_2-$, $-CF_2-$, $-C_2F_4-$, $-(C=O)O-$, $-O(C=O)-$, $-(C=O)NH-$, or $-NH(C=O)-$; and each of R^6 is independently H , CH_3 , or CF_3 . For example, the diamine compound can be 4,4'-oxydianiline (ODA), which has a chemical structure of

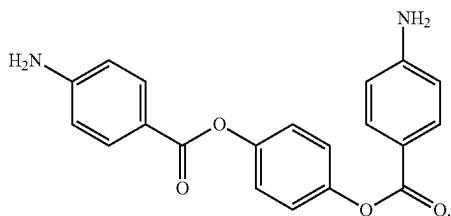


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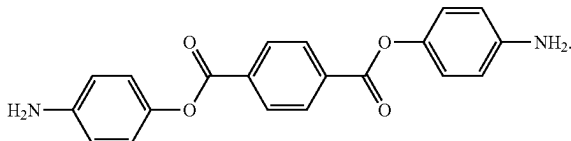
In some embodiments, the diamine compound can be 4-aminobenzoic acid 4-aminophenyl ester, which has a chemical structure of



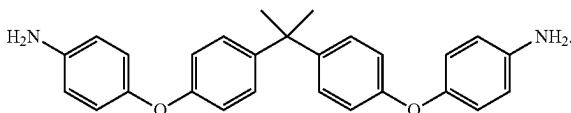
In some embodiments, the diamine compound can be 1,4-bis(4-aminophenoxy)benzene, which has a chemical structure of



In some embodiments, the diamine compound can be 1,4-benzenedicarboxylic acid bis(4-aminophenyl) ester, which has a chemical structure of



In some embodiments, the diamine compound can be 4,4'-(4,4'-isopropylidene diphenyl-1,1'-diylidioxy)dianiline, which has a chemical structure of



One embodiment of the disclosure provides a thin film, including the described polyimide. The thin film has a thickness of 10 μm to 75 μm , and a dielectric constant of 2.5 to 3 at 10 GHz. The dielectric constant of the polyimide thin film is related to the thickness of the polyimide thin film. If the dielectric constant of the polyimide thin film at 10 GHz is too high, the polyimide thin film cannot meet the requirements for its application as 5G board material.

Below, exemplary embodiments will be described in detail so as to be easily realized by a person having ordinary knowledge in the art. The inventive concept may be embod-

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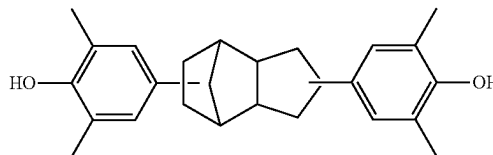
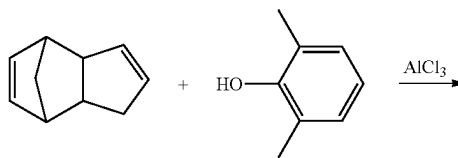
ied in various forms without being limited to the exemplary embodiments set forth herein. Descriptions of well-known parts are omitted for clarity.

EXAMPLES

In the following Examples, the dielectric constant of the thin film at 10 GHz was measured by the standard JISC2565, the dielectric loss of the thin film at 10 GHz was measured by the standard JISC2565, and the moisture absorption of the thin film was measured by the standard means of the follows. First, the film was heated in an oven to 105° C. for 30 mins, then moved to a drying box for cooling down and the weight of the film (W_0) was recorded. Next, the film was soaked in D. I. water for 24 hours and the weight after water absorption (W_1) was recorded. The moisture absorption of the thin film is $(W_1 - W_0)/W_0 \times 100\%$.

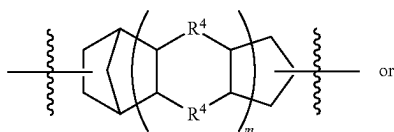
Example 1

AlCl_3 (0.10 mole, 13.11 g) serving as an acidic catalyst was added to and dissolved in 2,6-dimethylphenol (DMP, 1.18 mol, 144.15 g) solution, and dicyclopentadiene (DCPD, 0.60 mol, 78.00 g) was then added, and heated to react. After reaction, the solution was allowed to cool down to room temperature, and potassium hydroxide solution (5 wt %, 0.06 mol) was added to the reaction result to neutralize it. The diol compound was then extracted with toluene. The above reaction is shown below:

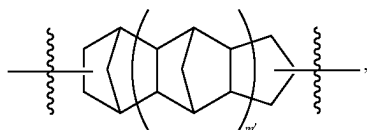


Pyridine (0.13 mol, 13.11 g) was added to trimellitic anhydride chloride (TMAC, 0.33 mol, 70.00 g) solution, which was then dropwise added to the diol compound solution to react for 24 hours. Afterward, the solvent of the reaction result was removed, and the reaction result was further dried in an oven to obtain an anhydride compound. The ^1H NMR spectrum of the anhydride compound is shown below: ^1H NMR (400 MHz, CDCl_3): δ 2.14 (s, 4H, aliphatic), 2.19 (s, 9H, aliphatic), 2.31 (s, 2H, aliphatic bridge), 7.14 (s, 4H, —Ar), 8.23 (d, 2H, phthalic anhydride, $J=7.88$ Hz), 8.76 (d, 2H, phthalic anhydride, $J=7.88$ Hz), 8.88 (s, 2H, phthalic anhydride). The reaction is shown below:

19

wherein R^1 is

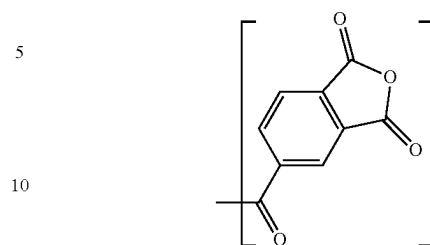
or



each of R^4 is independently C_{1-6} alkylene group, m is an integer of 0 to 10, and m' is an integer of 1 to 10; n is an integer of 1 to 10;

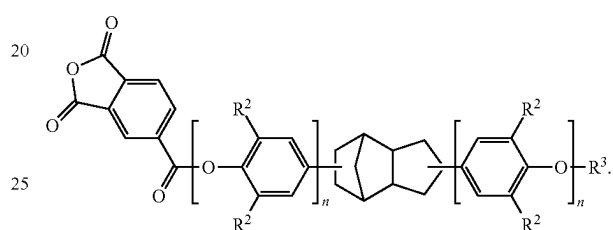
each of R^2 is independently hydrogen, saturated or unsaturated C_{1-6} hydrocarbon group, CF_3 , silanol group, silyl group, or $Al(OH)_3$; and

20

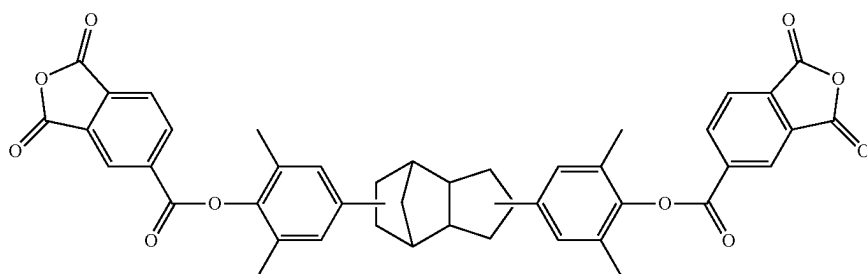
 R^3 is

silanol group, silyl group, or $Al(OH)_3$.

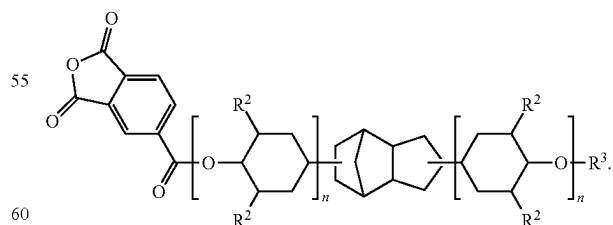
2. The anhydride compound as claimed in claim 1, having a chemical structure of



3. The anhydride compound as claimed in claim 1, having a chemical structure of

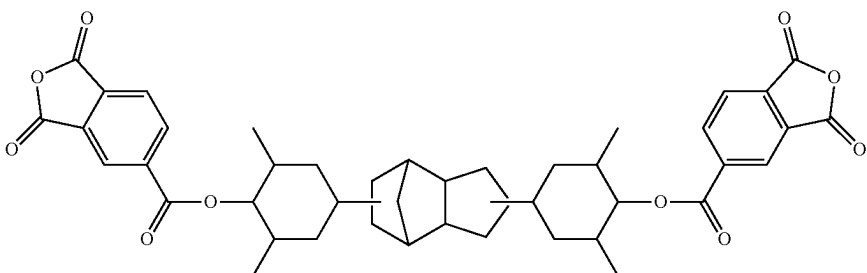


4. The anhydride compound as claimed in claim 1, having a chemical structure of



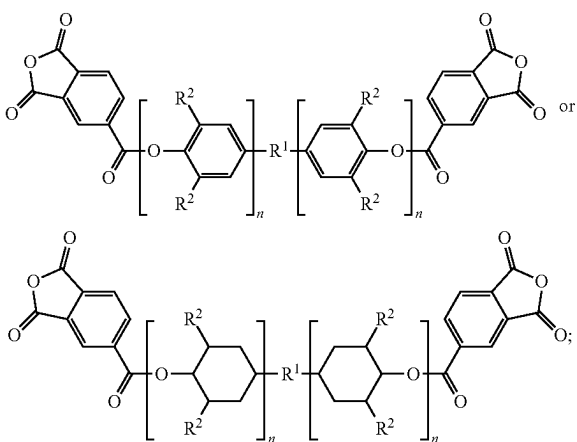
5. The anhydride compound as claimed in claim 1, having a chemical structure of

21

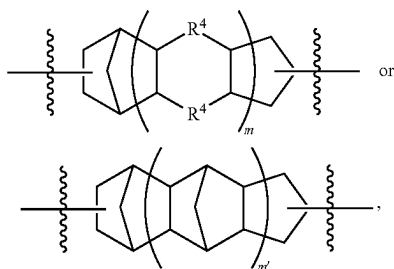


22

6. A polyimide, being formed by reacting an anhydride compound and a diamine compound;
wherein the anhydride compound has a chemical structure of

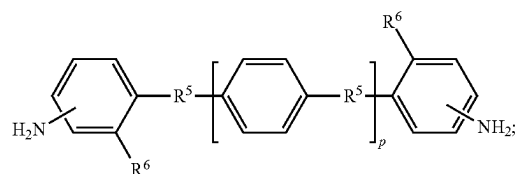


wherein R^1 is



each of R^4 is independently C_{1-6} alkylene group, m is an integer of 0 to 10, and m' is an integer of 1 to 10;
 n is an integer of 1 to 10; and
each of R^2 is independently hydrogen, saturated or unsaturated C_{1-6} hydrocarbon group, CF_3 , silanol group, silyl group, or $Al(OH)_3$.

7. The polyimide as claimed in claim 6, wherein the diamine compound has a chemical structure of

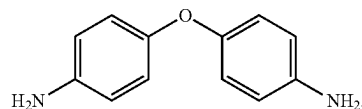


wherein p is an integer of 0 to 5;

each of R^5 is independently $-O-$, $-NH-$, C_{1-4} alkylene group, $-SO_2-$, $-CF_2-$, $-C_2F_4-$, $-(C=O)O-$, $-(C=O)-$, $-(C=O)NH-$, or $-NH(C=O)-$; and

each of R^6 is independently H, CH_3 , or CF_3 .

8. The polyimide as claimed in claim 6, wherein the diamine compound has a chemical structure of



9. A thin film, comprising the polyimide as claimed in claim 6, wherein the thin film has a thickness of 10 μm to 75 μm , and a dielectric constant of 2.5 to 3 at 10 GHz.

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